

Multi-Camera AR Navigation System for CT-Guided Needle Insertion Task

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ABSTRACT

In this work, we propose a multi-camera AR navigation system for the needle insertion task, including a high accuracy registration system and a user-friendly AR interface. Our proposed marker-based registration system utilizes an extended perception range and a high-precision calibration solution to address the difficulty of marker placement in clinical applications. The proposed AR interface also provides guidance on needle placement and insertion timing for needle insertion tasks. Experiments demonstrate that our proposed AR-guided system achieves clinically required insertion accuracy and facilitates needle-based insertion by improving needle placement accuracy, shortening completion time, and reducing radiation exposure.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods;

1 INTRODUCTION

Many medical treatments involve a critical step of inserting fine instruments like needles into the human body. The success of these treatments is almost dependent on the accuracy of the insertion. Inaccurate insertion may result in prolonged surgery time and potentially cause more pain. There has been an increasing interest in applying Augmented Reality (AR) technology to surgical navigation systems in recent years.

Accurate registration of AR guidance information on the patient is essential for AR navigation systems. Although some proposed registration systems employ marker-less methods, these methods are not fast or reliable enough for clinical applications [1]. The most commonly used methods today are marker-based methods and rely on optical [3] or planar markers [2]. Such marker-based approach estimates the camera pose for AR fusion by attaching markers to the patient and keeping them within the line of sight. As each surgical condition is different, marker placement and registration need to be readjusted prior to navigation, which complicates the surgical procedure and increases procedural time. Furthermore, as these markers are often placed near the surgical field, they may occupy the surgeon's field of view and interfere with the operation.

In this work, we propose a multi-camera hardware design to extend the field of view (FOV) of an AR device instead of using a single camera. We can take advantage of the extended FOV to place markers elsewhere in the operation room, such as the ceiling. Such markers are not 'visible' in the guidance interface to avoid interfering with the surgeon's operation. Also, since the position of these markers relative to the CT scanner is fixed, we only need to calibrate the registration system once, and do not need to adjust it before each operation. Although this registration system complicates the coordinate system transformations and may lead to the accumulation of AR fusion errors, we propose a joint optimization framework to obtain optimal results step by step and finally perform a global optimization to significantly reduce the AR fusion errors.

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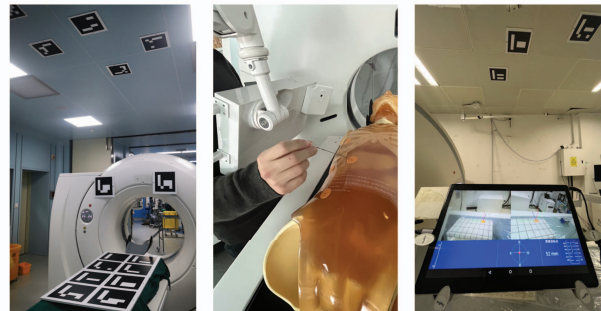


Figure 1: Left: Pre-operative setting; Mid: Intra-operative setting; Right: Phantom experiment

In addition, we introduce a user-friendly AR interface to better assist with the insertion task. We provide guidance for needle placement by tracking the needle position in real time and comparing it to the planned path. We also provide recommendations for optimal insertion times by monitoring the breathing cycle to reduce the impact of breathing on insertion accuracy. Finally, we conducted several phantom studies to demonstrate the robustness and reliability of our proposed system.

2 METHODOLOGY

2.1 Multi-camera collaborative registration system

The registration system includes a series of coordinate system transformation processes. Its purpose is to accurately superimpose AR navigation information on the patient. In other words, the guidance information in the CT coordinate system will be transformed into the coordinate system of the AR camera. We placed planar markers of known size in the operating room, and established a marker coordinate system through environment mapping techniques. A marker board placed on the operating table can be used to calculate the transformation between CT and marker coordinate system. Then, the transformation from the marker coordinate system to the AR tablet coordinate system can be estimated by tracking the planar markers. The final transformation from the tablet coordinate system to the camera coordinate system can be obtained through multi-camera calibration.

It is worth noting that the residual error will accumulate in multiple spatial transformations, thus affecting the accuracy of the final AR fusion. To address this issue, we propose a four-level hierarchical optimization framework. The bottom level is to optimize the pose estimation of individual markers in each captured frame, in which a weighting mechanism based on marker area is introduced to alleviate the influence of outliers on optimization.

In the second level, we connect and optimize the marker graph under normal constraints. We found that if the initial node is randomly selected, markers that are farther away from this node will have larger errors than those that are closer due to error accumulation. To solve this problem, we propose an effective way to automatically segment the graph. Each sub-graph will form a board object.

The 3D structure of the marker map is refined by minimizing the re-projection error in units of board objects.

In the third level, we exploit the rigid property between cameras combined with multi-camera calibration to further optimize the accuracy of the 3D structure of the marker map. The final step is to refine the entire system, including intrinsic parameters, camera poses, and the marker map by minimizing the following re-projection error cost function in all frames:

$$\min_{P_{c_{ref}}^c, P_b^c, P_m^b, K} \sum_{i=1}^C \sum_{t=1}^T \sum_{j=1}^M \omega_A \left[\omega_R \left\| {}^t x_{m_j} \right\| \right. \\ \left. - \psi \left(P_{c_{ref}}^c {}^t P_b^c P_m^b X_{m_j}, K \right) \right] + \omega_O \left\| V_{m_j} - N \left(P_m^b \right) \right\| \quad (1)$$

where K is the intrinsic parameter of camera. $P_{c_{ref}}^c$ is the extrinsic among cameras. P_b^c is the pose estimation from the marker board to the camera. P_m^b is the transformation matrix from the single marker to the board unit. $\psi(\cdot)$ is the standard camera projection function. x_{m_j} and X_{m_j} are the marker corners' 2D projection and its 3D position. V_{m_j} is the normal vector of marker and $N(\cdot)$ is the function to calculate the marker normal vector. ω_A, ω_R , and ω_O are weights for marker area, re-projection error and normal error. C, T and M are respectively the number of cameras, frames and markers.

Overall, our hierarchical calibration technique provides near-convergent initialization leading to reliable and accurate results.

2.2 Multi-dimensional guidance system

In order to effectively assist surgeons in operation, we improved the AR interface based on surgeons' clinical suggestions, as shown in Fig. 1. First, to help the surgeon locate the needle entry site, we designed a circular marker to identify it. The intended insertion point is located at its center.

Second, we need to further help surgeons adjust the needle placement, which requires needle tracking. We use a canny edge detector to extract possible needle regions, combined with depth-stopped color tracking, and finally utilize triangulated dual-camera results to filter and recover the needle's position and orientation in 3D space. The result will be visualized in the form of a cross-hair for the user's ease of understanding.

In addition, we employ a respiratory gating technique to provide waveform of respiratory data. It is worth noting that the breathing state during preoperative CT image acquisition will be automatically recorded as a reference line for subsequent operations.

3 EXPERIMENT

We conducted several experiments on synthetic and real data sets to systematically evaluate the performance of our method and compare it with state-of-the-art methods. The experimental equipment includes a computer (Ubuntu 18.04, Intel i5-12500H CPU), a low-cost tablet (MT6797 CPU) with two external USB cameras (HBVCAM-5M2134), a CT scanner (Lonwin), needles (15cm) and some medical consumables.

The first task was to evaluate the quality of the registration system, which consists of three main steps: environment mapping, multi-camera calibration and finally AR fusion. We first generated a synthetic dataset using Blender software to simulate the operating room and pre-operative operations. Based on the Blender dataset, we compare our method with those provided by the MarkerMapper [4] and MC-Calib [5] tools. Table 1 show that our method achieves the best results in terms of absolute angular error (ACE), absolute trajectory error (ATE), rotational error (RE), translation error (TE) and re-projection error in final AR fusion.

We also evaluated the AR fusion accuracy of our proposed registry in a clinical setting. Firstly, multiple needle insertion entries are simulated in the CT scan results of the human body model. User

Table 1: Comparison of registration accuracy

	Environment Mapping		Camera Calibration		AR Fusion
	ACE	ATE	RE	TE	Re-projection Error
MarkerMapper	0.113 m	0.043 m	1.3	0.020 m	4.91 px
MC-Calib	0.092 m	0.035 m	0.52	0.011 m	2.20 px
Our method	0.029 m	0.012 m	0.25	0.009 m	0.88 px

located the entry sites through our navigation system and the single-view AR system [2], and confirmed the distance error through the CT re-scan. The distance error of our method is 2.2 mm, which is close to the 1.8 mm result of the single-view AR system. This shows that our method can still achieve competitive calibration accuracy while reducing constraints on marker placement.

The second task was to evaluate the effectiveness of our AR-guided system for assisting needle insertion. We made a self-made phantom to simulate an abdominal puncture experiment. Multiple participants performed the needle insertion under the guidance of three different navigation systems. Our proposed method achieves the best first-time insertion accuracy of 3.76 ± 1.04 mm and the least average manipulation time of 20 ± 2.48 seconds, compared to the results of the conventional CT-guided method (8.39 ± 3.06 mm and 30 ± 2.45 seconds) and single-view AR systems [2] (4.39 ± 1.36 mm and 40 ± 2.69 seconds). This result demonstrates the robustness of the registration system and the feasibility of our guidance interface.

4 CONCLUSION

In this paper, we propose a multi-camera AR navigation system, including a multi-camera registration system and a user-friendly AR interface, to assist surgeons in needle insertion task. Our proposed system achieves higher accuracy than conventional configurations and simplifies the extra operations in other AR systems as much as possible. In the future, we will explore the potential value of the proposed method in more clinical practice.

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